Chemotherapy for bone cancer

You may have chemotherapy as part of your treatment for primary bone cancer. This uses anti-cancer drugs to destroy cancer cells.

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About chemotherapy for bone cancer

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy cancer cells.

Chemotherapy is an important part of treatment for most people with:

- osteosarcoma
- Ewing sarcoma
- spindle cell sarcoma.

It is not used often for other types of primary bone cancer.

Chemotherapy may be given:

- before you have surgery or radiotherapy
- after surgery or radiotherapy
- if the cancer comes back after treatment.

If you have osteosarcoma or Ewing sarcoma, you usually have chemotherapy before and after surgery.

Before having chemotherapy, you have tests to check your general health. It takes a few days to get the results. Your cancer doctor will talk with you about this.

Having chemotherapy

The drugs you have can depend on the type of bone sarcoma. You usually have a combination of drugs. These may include:

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- doxorubicin
- cisplatin
- methotrexate
- ifosfamide
- etoposide
- vincristine
- dactinomycin
- cyclophosphamide.

Combination chemotherapy for bone sarcoma

Primary bone cancer might be treated with a combination of chemotherapy drugs.

Osteosarcoma is usually treated with:

- MAP – methotrexate, doxorubicin and cisplatin
- AP – doxorubicin and cisplatin.

Children, teenagers and young adults (aged 30 or under) with osteosarcoma may be given a targeted therapy drug with the chemotherapy.

Ewing sarcoma is often treated with:

- VDC/IE – vincristine, doxorubicin and cyclophosphamide, alternating with ifosfamide and etoposide.
- We have more information about individual chemotherapy drugs.

How chemotherapy is given

Your cancer doctor or specialist nurse will tell you which drugs you will be given.

Chemotherapy drugs are usually given by injection into a vein (intravenously). They are given as a session called a cycle of treatment. You may have chemotherapy as an outpatient, or you may need to go into hospital for a few days.

Each treatment is followed by a rest period of a few weeks. This is to allow you to recover from any side effects.

The number of cycles you have depends on the type of primary bone cancer you have and how it responds to the treatment. There are different ways chemotherapy drugs given into a vein (intravenously) can be given. We have more information about the different ways chemotherapy may be given into the vein.

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Getting support

Macmillan is here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on 0808 808 00 00.
- Chat to our specialists online.
- Visit our chemotherapy forum to talk with people who have been affected by bone cancer, share your experience, and ask an expert your questions.

About our information

References

Below is a sample of the sources used in our information about primary bone cancer. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

European Society for Medical Oncology, ESMO. 2021. Bone sarcomas: ESMOeEURACANeGENTURISeERN PaedCan Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology. S. J. Strauss1 et al. Available at: https://www.esmo.org/guidelines/guidelines-by-topic/sarcoma-and-gist/bone-sarcomas (accessed July 2023)

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Reviewers

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor Fiona Cowie, Consultant Clinical Oncologist.

Our cancer information has been awarded the PIF TICK. Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

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The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

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Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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